

Mystic Motors — Economy Balancing Walkthrough

Full analysis of how shop prices, crate rewards, race rewards, and progression timers interconnect — and whether the 175-hour endgame target still holds.

1. The Currency Hierarchy

Currency	Source	Spent On	Can Be Bought?
Gems (hard)	IAP, ads, events	Time skips, boosters, crate purchases	Yes (IAP)
Coins (soft)	Races, crates	Car upgrades	Not directly
Spell Shards (soft)	Races, crates	Spell research	Not directly
Mastery XP	Races only	Tier gating (T2→T5)	Cannot be skipped

[!IMPORTANT]
Mastery XP is the master clock. It cannot be purchased, skipped, or boosted beyond 2x. This ensures a minimum play time regardless of spending.

2. Shop Prices — Everything Side by Side

2.1 Gem Packs (IAP)

Pack	Price	Gems	\$/Gem	Bonus
Handful	\$0.99	80	\$0.012	—
Pouch	\$4.99	500	\$0.010	+25%
Chest	\$9.99	1,200	\$0.008	+50%
Vault	\$19.99	2,800	\$0.007	+75%
Treasury	\$49.99	7,500	\$0.007	+88%
Hoard	\$99.99	16,000	\$0.006	+100%

Anchor: 1 gem ≈ \$0.01 at base rate, ~\$0.006 at best rate.

2.2 Time Skips (Direct Purchase)

Duration	Gems	Rate (gems/hr)	Real \$
5 min	5	60	\$0.05
15 min	8	32	\$0.08
1 hour	20	20 (base)	\$0.20
3 hours	50	16.7	\$0.50
8 hours	120	15	\$1.20
24 hours	300	12.5	\$3.00

3 days	700	9.7	\$7.00
7 days	1,400	8.3	\$14.00

Volume discount up to 58% off base for longer durations.

2.3 Boosters (Direct Purchase)

Booster	6hr	12hr	24hr	7 day
Coin 2×	80	130	200	1,000
Shard 2×	80	130	200	1,000
XP 2×	60	100	160	800

Boosters are deliberately underpriced (~60-80% of raw gem value) to feel like the "best deal" in the shop.

2.4 Crate Shop Prices (Buy Outright)

Rarity	Skip	Cost	Shop Price	Ratio
Common	20 gems	50 gems	2.5×	
Rare	50 gems	150 gems	3.0×	
Exotic	120 gems	400 gems	3.3×	
Legendary	200 gems	1,000 gems	5.0×	
Mythical	300 gems	2,500 gems	8.3×	

Shop price » skip price because buying outright skips both **earning** (racing) AND **waiting** (timer).

3. Crate Reward System — What You Get

3.1 Every Crate Always Gives (Guaranteed Base)

Reward	How It Works
Coins	Rank-scaled: $\text{COIN_CAPS}[\text{rank}][0] \times \text{tierMultiplier} \times (0.8-1.2)$
Spell Shards	Rank-scaled: $\text{shardBase}(\text{rank}) \times \text{tierMultiplier} \times (0.8-1.2)$
3 Cosmetic Items	Random cosmetics matching crate rarity

3.2 Guaranteed Coin Rewards (by rank and crate rarity)

Crate	Unranked	Silver III	Gold III	Platinum III	Diamond III	Hypersonic III
Common	400-600	780-1,170	1,080-1,620	1,500-2,250	2,100-3,150	8,260-12,390
Rare	800-1,200	1,560-2,340	2,160-3,240	3,000-4,500	4,200-6,300	16,520-24,780
Exotic	1,600-2,400	3,120-4,680	4,320-6,480	6,000-9,000	8,400-12,600	33,040-49,560
Legendary	3,200-4,800	6,240-9,360	8,640-12,960	12,000-18,000	16,800-25,200	66,080-99,120
Mythical	6,400-9,600	12,480-18,720	17,280-25,920	24,000-36,000	33,600-50,400	132,160-198,240

3.3 Guaranteed Shard Rewards

Shards scale **5×** across ranks (vs coins' **20×**) — intentionally much slower.

Crate	Unranked	Gold III	Diamond III	Hypersonic III
Common	1-2	2-4	3-5	4-6

Rare	3-5	5-8	6-10	8-12
Exotic	6-9	10-15	12-18	15-22
Legendary	10-15	16-24	20-30	25-37
Mythical	16-24	25-37	32-48	40-60

3.4 Bonus Utility Items (Multi-Slot Rolls)

On top of coins + shards + cosmetics, every crate rolls **1-3 utility items** (boosters and time skips):

Crate	Avg Items	Pool Contents
Common	1.28	5m/15m skips, 1hr boosters
Rare	1.41	15m/1hr skips, 1hr/6hr boosters
Exotic	1.55	1hr/3hr/8hr skips, 6hr/12hr boosters
Legendary	1.70	3hr/8hr/24hr skips, 12hr/24hr boosters, rare 7d
Mythical	1.85	8hr/24hr/3d/7d skips, 24hr/7d boosters

4. Gem-EV Analysis — Is Buying Crates Worth It?

This is the arbitrage check: can players game the system by buying crates to get cheap boosters/skips?

4.1 Expected Gem Value of Utility Items Per Crate

Crate	Avg Utility Items	Avg Gem-Value Per Item	Total EV	Shop Price
Common	1.28	~8 gems	~10 gems	50 gems
Rare	1.41	~18 gems	~25 gems	150 gems
Exotic	1.55	~40 gems	~62 gems	400 gems
Legendary	1.70	~80 gems	~136 gems	1,000 gems
Mythical	1.85	~150 gems	~278 gems	2,500 gems

4.2 EV Ratios

Crate	EV / Shop Price	EV / Skip Cost	Verdict
Common	20%	50%	No exploit
Rare	17%	50%	No exploit
Exotic	16%	52%	No exploit
Legendary	14%	68%	Safe (skip feels fair)
Mythical	11%	93%	Skip feels worthwhile

[!TIP]

EV/Shop price is 11-20% — buying crates for utility items is dramatically worse than buying items directly. The value proposition of crate purchases is the **cosmetics + excitement**, not utility grinding.

4.3 Why This Works

- **Booster durations scale with rarity.** Common crates only give 1hr boosters (worth ~18 gems). Legendary/Mythical give 24hr and 7d boosters. A player can't farm Common crates for cheap 24hr

boosters.

- **Coins are not in the utility pools.** They're guaranteed base, so the utility pool only contains boosters and time skips — fewer items = lower total EV.
- **Rate limiting.** Crate acquisition is gated by race wins (62.8% chance per race), opening timers (1-24hr), and the 4-slot cap. You can't mass-farm them.

5. Race Reward Scaling (Context for Crate Value)

5.1 Per-Race Earnings (1st Place)

Rank	Coins	XP	Shards	Mastery XP
Unranked	2,000	120	6	~127
Silver III	3,900	152	11	~159
Gold III	5,400	175	14	~182
Platinum III	7,500	199	17	~206
Diamond III	10,500	222	19	~229
Champion III	20,900	246	25	~254
Hypersonic III	41,300	250	30	~257

5.2 Crate Rewards as "Equivalent Races"

How many races-worth of rewards does each crate give?

Crate	Coin Equivalent	Shard Equivalent	Notes
Common	~0.25 races	~0.3 races	Pocket change — opens fast, low value
Rare	~0.5 races	~0.6 races	Decent top-up
Exotic	~1.0 race	~1.1 races	A full race's worth
Legendary	~2.0 races	~1.8 races	Very rewarding
Mythical	~4.0 races	~3.2 races	Jackpot — but 2% drop rate

[!NOTE]

A player earning ~0.63 crates/race (62.8% drop rate), weighted toward Common, gets an average of **~0.35 extra races worth of coins and ~0.4 races of shards** from crates per actual race played. This is a ~35-40% boost to earning rate.

6. Progression Impact — Does 175 Hours Still Hold?

6.1 Base Progression (No Crates, No Boosters)

Milestone	Races	Gameplay Hours
Tier 2 (Rank 5)	191	~10 hr
Tier 3 (Rank 10)	732	~37 hr
Tier 4 (Rank 20)	1,975	~99 hr
Tier 5 (Rank 30)	3,503	~175 hr

6.2 With Crate Rewards (F2P)

Crates effectively give a ~35-40% boost to coin/shard income. But **mastery XP is the real gate**, and crates don't give mastery XP.

What Crates Accelerate	Impact
Coins (car upgrades)	~35% faster coin accumulation → reach upgrade costs sooner
Shards (spell research)	~40% faster shard accumulation → can start research sooner
Upgrade timers (time skips from crates)	~2-5 hours saved over 175 hr for a F2P player
Mastery XP	No effect — still requires same number of races

6.3 Revised Effective Timeline

Profile	Mastery Gate	Progression Bottleneck	Effective Hours to Tier 5
F2P (no spending)	175 hr	Mastery XP (can't skip)	~175 hr (unchanged)
F2P with crate skips	175 hr	Mastery XP	~175 hr (timer savings: ~2-5hr)
Light spender (\$20)	175 hr	Mastery XP	~170 hr (skip some timers + buy a few boosters)
Moderate spender (\$50-100)	175 hr	Mastery XP	~165 hr (XP booster compresses races)
Whale (\$200+)	175 hr	Mastery XP	~155 hr (max XP booster uptime)

[!IMPORTANT]

The 175-hour target holds because mastery is the bottleneck, not coins or shards. Crates and spending compress the *secondary* gates (upgrade timers, resource gathering) but cannot skip the primary gate. Even a whale cannot reach Tier 5 in under ~150 hours.

6.4 Where Crates Actually Help Most

Phase	Hours	Crate Impact
Early game (0-10hr)	Player is coin-starved, Tier 1 upgrades are fast	Crate coins help buy early upgrades faster
Mid game (10-37hr)	Timer waits get meaningful (1-4hr), shards start mattering	Time skips from crates save 15-60min of waiting
Late-mid (37-99hr)	Tier 3-4 upgrades cost 50k-120k coins, spell research needs hundreds of shards	Rank-scaled crate coins are now significant (5k-15k per Legendary crate)
Endgame (99-175hr)	Tier 5 upgrades cost 293k coins per car, spells need 2,000 shards for Lv5	Mythical crate can give 30k-50k coins — meaningful but not game-breaking

6.5 Is 175 Hours Too Long?

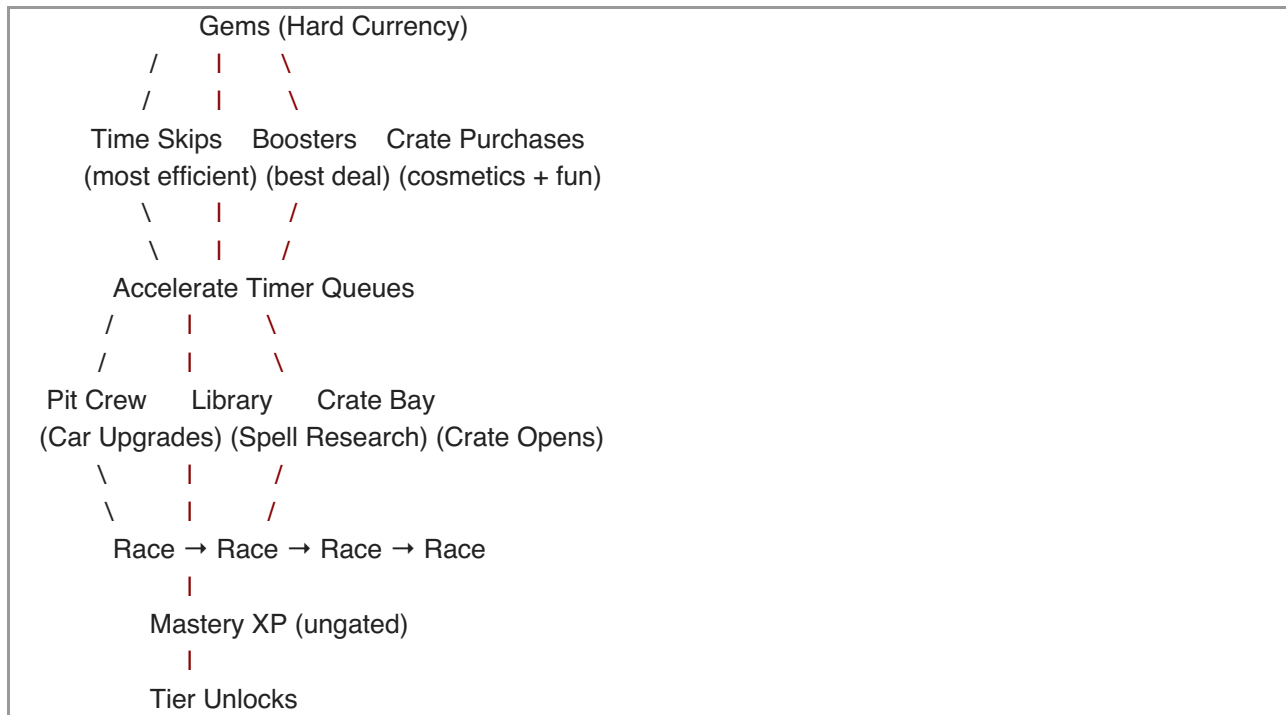
The 175hr figure is **calendar time**, not wall-clock gaming. At 10 races/day (~30 min/day):

Metric	Value
Real-world days to Tier 5	~350 days (~1 year)
Actual gameplay hours	~175 hours
Daily play session	~30 minutes
Comparison: Clash Royale to max	~3-4 years F2P

Comparison: Brawl Stars to max ~2 years F2P

175 hours of gameplay spread over ~1 year is actually **moderate** for a F2P mobile game. The crate system ensures players always have something to look forward to during this journey.

7. Summary — The Balancing Pyramid



Key insight: Gems accelerate the *secondary* systems (timers, resources), but the *primary* system (mastery) can only be progressed by playing races. This ensures engagement without allowing pay-to-win shortcuts.

8. Guardrails & Anti-Exploit Summary

Risk	Protection
Crate farming for cheap boosters	EV/shop price = 11-20% → direct purchase is always better
Smurf lower ranks for easy wins	Rank-scaled rewards → higher rank 6th > lower rank 1st
Skip all timers with gems	Mastery still requires the same # of races
Hoard 7-day boosters from crates	Only available in Legendary (3.3% drop) / Mythical (2% drop) pools
Spam open Common crates for shards	Common crate shards = 1-2 (Unranked) to 4-6 (Hypersonic) — trivial